

1 Phase 1: Comprehensive Research & Literature Review (Weeks 1-6)

1.1 Objectives

- Conduct an in-depth literature review to identify gaps and refine research questions.
- Analyze existing solutions and theoretical frameworks.
- Gather foundational knowledge on technologies without committing to implementation yet.
- Produce initial writing outputs to build thesis structure.

1.2 Research Tasks

1.2.1 Weeks 1-2: Literature Search & Review

- Conduct systematic literature review (expand bibliography to 50+ papers).
- Analyze competing solutions (Ethereum-based, centralized systems).
- Study Polkadot architecture, parachain mechanics, ink! smart contracts, and XCM protocol specifications.
- Review cross-chain implementations and content rights management literature.

1.2.2 Weeks 3-4: Deep Dive & Gap Analysis

- Examine blockchain-based content monetization models (e.g., NFTs, tokenization).
- Analyze interoperability standards and smart contract platforms.
- Identify research gaps and potential redesign opportunities.
- Create an annotated bibliography and literature review matrix.

1.2.3 Weeks 5-6: Theoretical Framework Development

- Refine research questions and objectives based on findings.
- Outline high-level system concepts (without detailed architecture).
- Document potential challenges in cross-chain rights management.

1.3 Writing Tasks (Target: 4,000 words)

- Introduction draft (1,000 words).
- Background and Related Work chapter (2,000 words).
- Research Questions and Objectives chapter (1,000 words).
- Create annotated bibliography.

1.4 Deliverables

- Literature review matrix (50+ papers).
- Annotated bibliography.
- Initial chapters: Introduction, Background and Related Work, Research Questions and Objectives (drafts).
- Research findings report for potential pivots in technical approach.